AA411

Pilot Guidebook Test Plan

Team Name:	ROFASO
Project Name:	Damage Assessment Protocol (D.A.P.)
Sub-system Name:	Documentation
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1 Objective of Test Plan

The goal of this plan is to test that the Pilot Guidebook can successfully guide a pilot through operation of the Damage Assessment Protocol (D.A.P.) system and allow the pilot to correctly interpret all outputs generated by the system. The pilot will use the Pilot Guidebook to execute the D.A.P. workflow, generate a Final Damage Assessment Report, identify the Primary Diagnosis, understand the reported fault severity, and explain how the diagnosis was generated.
The test also verifies that the Pilot Guidebook provides sufficient information for pilots to understand percent error, window scores, fault thresholds, fault confidence, and the relationship between the Digital Twin and Diagnoser.

1.1 i Scope

- Function: Pilot Guidebook
- Type: Documentation Verification
- Outputs:
  - Successful D.A.P. workflow execution
  - Recorded Primary Diagnosis
  - Recorded fault severity
  - Interpretation of D.A.P. outputs
  - Understanding of diagnosis generation process

2 Test Parameters and Variables

2.1 i Parameters

Parameters	**Description**	**Value**	**Source**
Pilot Guidebook	Documentation used by pilot during testing	Current Revision	Documentation Team
Mission Data	Seaglider mission data used for testing	All	SG.github
D.A.P. Software	Damage Assessment Protocol software	Current Revision	D.A.P. Team
MATLAB Version	MATLAB environment used to run D.A.P.	R2024b	MATLAB

Table 1: Pilot Guidebook Validation Parameters

2.2 ii Variables

Variable	**Description**	**Units**
	Independent Variables (Control Inputs)	
P1940008	Mission dataset containing a 25% wing fault	Mission data units
P1940009	Mission dataset containing a nominal case	Mission data units
P1950701	Mission dataset containing a 50% rudder fault	Mission data units
P1950702	Mission dataset containing a 50% wing fault	Mission data units
	Dependent Variables (State Variables)	
Primary Diagnosis	Fault case identified by D.A.P.	Classification
Fault Severity	Reported fault severity	Percent
Percent Error	Difference between mission and Digital Twin outputs	Percent
Window Score	Window-based fault detection score	Score
Fault Confidence	Confidence associated with identified fault	Percent
Pilot Interpretation	Ability to correctly explain D.A.P. outputs	Pass/Fail
Workflow Completion	Ability to execute the D.A.P. workflow using only the Pilot Guidebook	Pass/Fail

Table 2: Pilot Guidebook Validation Variables

3 Test Matrix

Full Test Matrix:

Test ID	**Mission Dataset**	**Expected Diagnosis**
PG1	P1940008	25% Wing Fault
PG2	P1940009	Nominal
PG3	P1950701	50% Rudder Fault
PG4	P1950702	50% Wing Fault

Table 3: Pilot Guidebook Validation Test Matrix

4 Traceability

Requirement PG-1 is verified by test cases PG1 through PG4.
True/False? ______

Requirement PG-2 is verified by test cases PG1 through PG4.
True/False? ______